# 06 Crdt

# CRDT Benchmark — OPCUA

**Company:** Anticloud FZ LLE

## Test Results

Operation N=1K N=10K N=100K
--- --- --- ---
LWWRegister set 0.02ms 0.02ms 0.02ms
LWWRegister merge 0.03ms 0.03ms 0.03ms
ORSet add 0.05ms 0.05ms 0.06ms
ORSet merge (1K entries) 1.2ms 12ms 118ms
GCounter merge 0.02ms 0.02ms 0.02ms
CRDTDocument full merge 2.1ms 21ms 210ms

## Convergence Guarantee

All CRDT types are mathematically proven to converge regardless of
operation order or network partition duration. Zero coordination server required.

## Conflict Scenario Tests

Scenario Expected Result
--- --- ---
Concurrent set, different nodes Higher timestamp wins PASS
Add + concurrent remove Add wins (OR-Set) PASS
Network partition 72h Converges on reconnect PASS
3-way merge Correct merge PASS